Progress report

Project scope update

When the project was initially proposed, the goal was to examine the relationship between surprising Premier League match outcomes and social media engagement on official club X (Twitter) accounts by identifying the “final score” post, assuming it’s the most engaged match-related tweet within a post-match time window.

However, tests discovered that the most engaged post-match tweet does not necessarily target the “final score” post. Thus, instead of selecting a single tweet, the analysis now compares average engagement levels before and after each match as uses “engagement lift” (avg(post-match_tweets_engagement) - avg(pre-match_tweets_engagement)) to measure the influence of match results.

- The pre-match time window is defined as: Kickoff time − 6 hours → Kickoff
- The post-match time window is defined as: Kickoff time + 105 minutes → Kickoff time + 465 minutes

This refinement improves the robustness, reproducibility, and statistical defensibility of the project.

Data sources

Match Data – Match-level data for the 2025–2026 Premier League season is retrieved using the football-data.org API: https://www.football-data.org/

Betting Odds Data - Pre-match betting odds are automatically downloaded via direct CSV URL: https://www.football-data.co.uk/mmz4281/2526/E0.csv

X (Twitter) Engagement Data – Engagement data is retrieved using the official X API (v2) via the tweepy library. For each club, tweets are retrieved using the user ID (stored locally to avoid repeated username lookups). For each tweet, the following public metrics are collected:

- like_count
- reply_count
- retweet_count
- quote_count
- created_at

Issues / difficulties

A major difficulty encountered was reliably identifying match-related tweets. Clubs use highly variable language when posting final results posts, and the most engaged post-match post turns out to not necessarily be the final post. Additionally, exact final whistle times are not provided by the football-data API, and stoppage time varies across matches. This made precise identification of the “correct” post impractical. To address this, the methodology was redesigned to measure engagement lift using structured time windows rather than selecting individual match-result tweets.

Furthermore, the X API’s pay-per-resource model required careful design to minimize unnecessary pagination and repeated queries. Ensuring consistent UTC time handling across match data, betting odds, and tweet timestamps was also necessary to prevent misalignment across data sources.